Curtis Chong

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Google Scholar

Education

University of Waterloo · Bachelor of Software Engineering — Apr 2023

Works

TorchSim Maintainer

- Introduced support for mixed periodic boundary conditions and attribute tracking in SimState
- Resolved issues with batching, property predictions, and SimState management
- Simplified batching algorithms, added SimState validation, and typed variables to enhance dev experience

LitXBench · A Benchmark for Extracting Experiments in Papers — radical-ai.github.io/litxbench

- Demonstrated that LLMs perform best when experiments are extracted in code format, not JSON
- Revealed that code empowers LLMs to perform accurate calculations and semantically validate results

MATRIX · A Multimodal MatSci Benchmark and Post-Training Framework — arxiv.org/pdf/2602.00376

- Trained multi-modal LLMs and benchmarked their performance on materials discovery tasks
- Demonstrated that training LLMs to caption materials science figures greatly improves model performance
- Developed response alignment tools to calibrate LLM-as-a-judge scoring against human baselines

Extending the Platonic Transformers MLIP for Periodic Lattices — *paper in progress*

- Developed evals to measure the smoothness of potential energy surfaces — bit.ly/smoothness-evals
- Trained DPA3 and Crystalformer variants to identify mechanisms for potential energy surface smoothness

E3Simple · The Simplest E(3)-equivariant Graph NeuralNet library — bit.ly/e3simple

- Engineered the simplest-to-understand E(3)-GNN library from scratch in 500 lines of code
- Incorporated the best ideas of E3nn and E3x, providing clear documentation of tradeoffs

Experience

Radical AI — New York

ML Engineer · ML Team — Feb 2025 - Present

- Developed **panoptic segmentation models** to determine the volume fraction of alloy phases
- Trained models to predict alloy castability and identify surface defects (scratches, pores, cleaning residue)
- Mined property prediction heuristics from papers to build ensemble predictors that atomistics can't model
- Indexed compositions experimentally made in papers to determine the novelty of our experiments
- Developed **embedding models** for microscopy images to find alloys with similar microstructure
- Trained LLMs to predict a composition's crystal structure using supervised fine-tuning and RL

Citadel Securities — Chicago

SRE Intern · Trade Execution Services Team — Sep – Dec 2022

- Crafted high-performant scripts to efficiently extract and transform **terabytes of logs** into KDB
- Upgraded an XML interpreter to parse nested FlatBuffer messages, simplifying hundreds of lines of code

Jane Street — New York

SWE Intern · Cybersecurity Team — Jan – Apr 2022

- Developed filters to block network packets from malicious IPs or carry invalid TLS certificates
- Classified IPs as safe or malicious in under **20 µs** via a trie-based lookup table

BitGo — San Francisco

SWE Intern · Trade Execution Team — Jan – Apr 2020

- Wrote **trading algorithms** (Sweep and TWAP) to place orders across crypto exchanges at optimal prices
- Designed race-free state machines and exchange balance tracking services to ensure stable trades